

iYangians as coideal subalgebras of Yangians

Kang Lu

Based on joint works W. Wang and W. Zhang

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Yangians vs twisted Yangians

Yangian $\mathcal{Y}$

- Lie algebra $\mathfrak{g}$
- R-matrix
- Yang-Baxter equation
- Hopf algebra
- finite W-algebras of type A
- Affine Grassmannian slices
- ...

iYangian (twisted Yangian) ${}^i\mathcal{Y}$

- symmetric pair $(\mathfrak{g}, \mathfrak{g}^\theta)$
- K-matrix
- reflection equation
- coideal subalgebra of the Yangian
- finite W-algebras of classical type
- Affine Grassmannian i slices
- ...

Yangians in R-matrix and J presentations

R-matrix presentation [Faddeev school]

$\mathcal{Y}_{\mathcal{R}}$ is generated by elements encoded by the generating matrix $T(u)$ subject to

$$R(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R(u-v)$$

and possibly extra central relations.

$$\Delta(T(u)) = T_1(u)T_2(u), \quad S(T(u)) = T(u)^{-1}$$

J presentation [Drinfeld85]

$\mathcal{Y}_{\mathcal{J}}$ is generated by elements x and $J(x)$ for $x \in \mathfrak{g}$ subject to complicated relations.

$$\Delta(x) = x \otimes 1 + 1 \otimes x, \quad \Delta(J(x)) = J(x) \otimes 1 + 1 \otimes J(x) + \frac{1}{2}[x \otimes 1, \Omega_{\mathfrak{g}}].$$

Current algebra $\mathfrak{g}[z]$

- Let e_i, f_i, h_i for $i \in \mathbb{I}$ be Chevalley generators of $\mathfrak{g}$
- Cartan matrix $C = (c_{ij})_{i,j \in \mathbb{I}}$ (focusing on type ADE)
- Set $x_{i,r}^+ = e_i z^r, x_{i,r}^- = f_i z^r, h_{i,r} = h_i z^r$
- Current presentation:

$$[h_{i,r}, h_{j,s}] = 0 \quad [x_{i,r}^+, x_{j,s}^-] = \delta_{ij} h_{i,r+s}$$

$$[h_{i,0}, x_{j,s}^\pm] = \pm c_{ij} x_{j,s}^\pm$$

$$[h_{i,r+1}, x_{j,s}^\pm] - [h_{i,r}, x_{j,s+1}^\pm] = 0$$

$$[x_{i,r+1}^\pm, x_{j,s}^\pm] - [x_{i,r}^\pm, x_{j,s+1}^\pm] = 0$$

$$[x_{i,r}^\pm, x_{j,s}^\pm] = 0 \quad \text{if } c_{ij} = 0$$

$$\text{Sym}_{r_1, r_2} [x_{i,r_1}^\pm, [x_{i,r_2}^\pm, x_{j,s}^\pm]] = 0 \quad \text{if } c_{ij} = -1$$

Yangian $\mathcal{Y}$ in Drinfeld presentation

- $\{a, b\} := ab + ba$ and $\hbar$ a formal variable

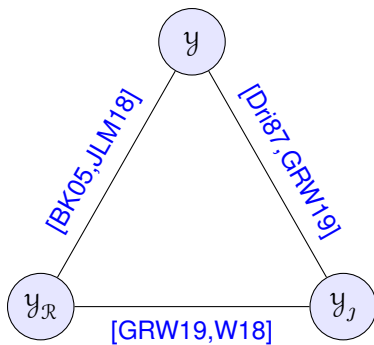
[Drinfeld'87] Drinfeld new/current presentation

$\mathcal{Y}$ is a $\mathbb{C}[\hbar]$ -algebra generated by $\xi_{i,r}, x_{i,r}^{\pm}$, $i \in \mathbb{I}$ and $r \in \mathbb{N}$ subject to

$$\begin{aligned} [\xi_{i,r}, \xi_{j,s}] &= 0 & [x_{i,r}^+, x_{j,s}^-] &= \delta_{ij} \xi_{i,r+s} \\ [\xi_{i,0}, x_{j,s}^{\pm}] &= \pm c_{ij} x_{j,s}^{\pm} \\ [\xi_{i,r+1}, x_{j,s}^{\pm}] - [\xi_{i,r}, x_{j,s+1}^{\pm}] &= \pm \frac{c_{ij}}{2} \hbar \{ \xi_{i,r}, x_{j,s}^{\pm} \} \\ [x_{i,r+1}^{\pm}, x_{j,s}^{\pm}] - [x_{i,r}^{\pm}, x_{j,s+1}^{\pm}] &= \pm \frac{c_{ij}}{2} \hbar \{ x_{i,r}^{\pm}, x_{j,s}^{\pm} \} \\ [x_{i,r}^{\pm}, x_{j,s}^{\pm}] &= 0 & \text{if } c_{ij} &= 0 \\ \text{Sym}_{r_1, r_2} [x_{i,r_1}^{\pm}, [x_{i,r_2}^{\pm}, x_{j,s}^{\pm}]] &= 0 & \text{if } c_{ij} &= -1 \end{aligned}$$

- Flat deformation of $\mathcal{U}(\mathfrak{g}[z])$, $\xi_{i,r} \rightsquigarrow h_i z^r$, $x_{i,r}^+ \rightsquigarrow e_i z^r$, $x_{i,r}^- \rightsquigarrow f_i z^r$ as $\hbar \rightarrow 0$

Equivalences



Twisted Yangians in R-matrix and J presentations

R-matrix presentation [Sklyanin88, Olshanski92]

${}^i\mathcal{Y}_{\mathcal{R}}$ is generated by elements encoded by the generating matrix $S(u)$ subject to

$$R(u-v)S_1(u)R(u+v)S_2(v) = S_2(v)R(u+v)S_1(u)R(u-v)$$

and some extra relations. ${}^i\mathcal{Y}_{\mathcal{R}}$ is a coideal subalgebra of $\mathcal{Y}_{\mathcal{R}}$: e.g. $S(u) = T^t(-u)KT(u)$.

- Let $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ be the eigenspace decomposition of an **arbitrary** involution θ .

J presentation [MacKay02]

${}^i\mathcal{Y}_J$ is the subalgebra of $\mathcal{Y}_J$ generated by all the elements

$$x, \quad B(y) := J(y) - \frac{1}{4}[y, C_{\mathfrak{k}}], \quad \text{where } x \in \mathfrak{k}, y \in \mathfrak{p}.$$

${}^i\mathcal{Y}_J$ is a right coideal subalgebra: $\Delta(B(y)) = B(y) \otimes 1 + 1 \otimes B(y) - [1 \otimes y, \Omega_{\mathfrak{k}}]$.

- Idea of proving ${}^i\mathcal{Y}_{\mathcal{R}} \cong {}^i\mathcal{Y}_J$ is clear as both are known to be subalgebras of $\mathcal{Y}$

Twisted current algebras $\mathfrak{g}[z]^\theta$

- $\theta : \mathfrak{g} \rightarrow \mathfrak{g}, \quad e_i \mapsto -f_i, \quad f_i \mapsto -e_i, \quad h_i \mapsto -h_i$
- Extend θ to $\mathfrak{g}[z]$ by

$$\theta(xz^r) = \theta(x)(-z)^r$$

- The twisted current algebra $\mathfrak{g}[z]^\theta$ is generated by

$$x_{i,r} := ((-1)^r f_i - e_i)z^r, \quad t_{i,2r+1} = 2h_i z^{2r+1}$$

- Current presentation:

$$[t_{i,r}, t_{j,s}] = 0, \quad t_{i,2r} = 0$$

$$[t_{i,r+1}, x_{j,s}] - [t_{i,r-1}, x_{j,s+2}] = 0$$

$$[x_{i,r+1}, x_{j,s}] - [x_{i,r}, x_{j,s+1}] = -2\delta_{ij}(-1)^r t_{i,r+s+1}$$

$$[x_{i,r}, x_{j,s}] = 0 \quad \text{if } c_{ij} = 0$$

$$\text{Sym}_{k_1, k_2} [x_{i,k_1}, [x_{i,k_2}, x_{j,r}]] = 2(-1)^{k_1} x_{i,k_1+k_2+r} \quad \text{if } c_{ij} = -1$$

iYangian (Twisted Yangian in Drinfeld presentation)

- Recall $\{a, b\} = ab + ba$ and $\hbar$ a formal parameter

Definition [L-Wang-Zhang23-24]

The $\mathbb{C}[\hbar]$ -algebra ${}^i\mathcal{Y}$ (**split type**) is generated $h_{i,r}, b_{i,s}$ for $i \in \mathbb{I}, r, s \in \mathbb{N}$ and subject to

$$[h_{i,r}, h_{j,s}] = 0, \quad h_{i,2r} = 0$$

$$[h_{i,r+1}, b_{j,s}] - [h_{i,r-1}, b_{j,s+2}] = c_{ij}\hbar\{h_{i,r-1}, b_{j,s+1}\} + \frac{c_{ij}^2\hbar^2}{4}[h_{i,r-1}, b_{j,s}]$$

$$[b_{i,r+1}, b_{j,s}] - [b_{i,r}, b_{j,s+1}] = \frac{c_{ij}\hbar}{2}\{b_{i,r}, b_{j,s}\} - 2\delta_{ij}(-1)^r h_{i,r+s+1}$$

$$[b_{i,r}, b_{j,s}] = 0 \quad \text{if } c_{ij} = 0$$

$$\text{Sym}_{k_1, k_2}[b_{i, k_1}, [b_{i, k_2}, b_{j, r}]] = (-1)^{k_1+1}[h_{i, k_1+k_2+1}, b_{j, r-1}] \quad \text{if } c_{ij} = -1$$

Flat deformation of twisted current algebra $\mathcal{U}(\mathfrak{g}[z]^\theta)$, $b_{i,r} \rightsquigarrow x_{i,r}$, $h_{i,r} \rightsquigarrow t_{i,r}$ as $\hbar \rightarrow 0$

Remarks

- The presentation of ${}^i\mathcal{Y}$ is obtained
 - (1) via Gauss decomposition of $S(u)$ for type A
 - (2) via degeneration of affine i quantum groups ${}^i\mathcal{U}$ in Drinfeld presentation established by [Ming Lu-Wang](#) (ADE) and [Zhang](#) (BCFG)
- For type A, ${}^i\mathcal{Y}$ is a coideal subalgebra of $\mathcal{Y}$ as ${}^i\mathcal{Y} \cong {}^i\mathcal{Y}_{\mathcal{R}}$ (type BCD was posted on arXiv by [Regelskis](#) last Thursday)
- For other types, it remains unclear from the degeneration procedure if ${}^i\mathcal{Y}$ is a coideal subalgebra of $\mathcal{Y}$
 1. The embedding ${}^i\mathcal{U}$ into $\mathcal{U}$ via **Drinfeld generators** is unknown
 2. **Compatibility of filtrations** on ${}^i\mathcal{U}$ and on $\mathcal{U}$ under the embedding ${}^i\mathcal{U} \hookrightarrow \mathcal{U}$ is unclear

Goal

We would like to address this question for other types.

Embedding into Yangian

Theorem [L25]

Assume that $\mathfrak{g}$ is not of type G_2 . ${}^i\mathcal{Y}$ is a subalgebra of $\mathcal{Y}$ via the following identification

$$h_{i,1} \mapsto 2\xi_{i,1} - \xi_i^2 + \sum_{\alpha \in \Phi^+} (\alpha, \alpha_i) (x_\alpha^+)^2$$

$$b_{i,0} \mapsto x_i^+ - x_i^-$$

$$b_{i,1} \mapsto x_{i,1}^+ + x_{i,1}^- + \frac{1}{2} \sum_{\alpha \in \Phi^+} \{[x_i^+, x_\alpha^+], x_\alpha^+\} - \frac{1}{2} \{x_i^+, \xi_i\}$$

Surprisingly, the formulas are similar to the identification of $\mathcal{Y}_j$ with $\mathcal{Y}$ [Drinfeld87]

$$J(\xi_i) \mapsto \xi_{i,1} - \frac{1}{2}\xi_i^2 + \frac{1}{4} \sum_{\alpha \in \Phi^+} (\alpha, \alpha_i) \{x_\alpha^+, x_\alpha^-\}$$

$$J(x_i^\pm) \mapsto x_{i,1}^\pm \pm \frac{1}{4} \sum_{\alpha \in \Phi^+} \{[x_i^\pm, x_\alpha^\pm], x_\alpha^\mp\} - \frac{1}{4} \{x_i^\pm, \xi_i\}$$

Minimalistic presentation

Theorem [L25]

If $\mathfrak{g}$ is not of type A_1 , $B_2 \cong C_2$, or G_2 , then ${}^v\mathcal{Y}$ is generated by $\{h_{i,1}, b_{i,0}, b_{i,1}\}_{i \in \mathbb{I}}$, subject to

$$[h_{i,1}, h_{j,1}] = 0$$

$$[h_{i,1}, b_{j,0}] = 2c_{ij}b_{j,1}$$

$$[b_{i,1}, b_{j,0}] - [b_{i,0}, b_{j,1}] = \frac{1}{2}c_{ij}\{b_{i,0}, b_{j,0}\} - 2\delta_{ij}h_{i,1}$$

$$[b_{i,0}, b_{j,0}] = 0 \quad (c_{ij} = 0)$$

$$[b_{i,0}, [b_{i,0}, b_{j,0}]] = -b_{j,0} \quad (c_{ij} = -1)$$

If $\mathfrak{g}$ is of type A_1 , $B_2 \cong C_2$, or G_2 , then an additional relation

$$\left[h_{i,1}, [b_{i,1}, [h_{i,1}, b_{i,1}]] \right] = 4[b_{i,1}^2, h_{i,1}] \quad (\Longleftrightarrow [h_{i,1}, h_{i,3}] = 0)$$

should be included for any **single** $i \in \mathbb{I}$.

Coideal structure

Corollary

Assume that $\mathfrak{g}$ is not of type G_2 .

1. ${}^i\mathcal{Y}$ is a right coideal subalgebra of $\mathcal{Y}$, i.e. $\Delta({}^i\mathcal{Y}) \subset {}^i\mathcal{Y} \otimes \mathcal{Y}$, and

$$\Delta(h_{i,1}) = h_{i,1} \otimes 1 + 1 \otimes h_{i,1} + 2 \sum_{\alpha \in \Phi^+} (\alpha, \alpha_i) (x_\alpha^+ - x_\alpha^-) \otimes x_\alpha^+.$$

2. The i Yangian ${}^i\mathcal{Y}$ is isomorphic to the twisted Yangian ${}^i\mathcal{Y}_j$ in the J presentation.

Thank you!

Estimation

- $Q_+ := \{ \sum_{i \in \mathbb{I}} k_i \alpha_i \mid k_i \in \mathbb{N}, \sum_{i \in \mathbb{I}} k_i > 0 \}$
- Set $h_i(u) = 1 + \sum_{r \geq 0} h_{i,2r+1} u^{-2r-2}$, $b_i(u) = \sum_{r \geq 0} b_{i,r} u^{-r-1}$

Theorem [L25]

Let $\mathfrak{g}$ be a simple Lie algebra which is not of type G_2 . We have

$$h_i(u) \equiv \xi_i(u) \xi_i(-u) \pmod{\mathcal{Y}_{Q_+}[[u^{-1}]]}$$

$$b_i(u) \equiv \frac{1}{2} \{ x_i^+(u), \xi_i(-u) \} + x_i^-(u) \pmod{\mathcal{Y}_{\alpha_i + Q_+}^{\geq 0}[[u^{-1}]]}$$

$$\Delta(h_i(u)) \equiv h_i(u) \otimes \xi_i(u) \xi_i(-u) \pmod{{}^i\mathcal{Y} \otimes \mathcal{Y}_{Q_+}[[u^{-1}]]}$$

$$\Delta(b_i(u)) \equiv b_i(u) \otimes \xi_i(-u) + 1 \otimes b_i(u) \pmod{{}^i\mathcal{Y} \otimes \mathcal{Y}_{Q_+}[[u^{-1}]]}$$

- Conjectured by [\[Wang-Zhang22\]](#) for affine iquantum groups
- Application: compute the ${}^i q$ -character of the restriction $\mathcal{Y}$ -module